

SPOORTHI M

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Education

Amrita School of Computing, Amrita Vishwa Vidyapeetham

Sep. 2021 – May 2025

Bachelor of Technology in Computer Science & Engineering

Bengaluru, India

CGPA: 9.09/10 | University Rank: 6th out of 200 students

Research Interests

Medical Image Analysis | Neural Radiance Fields (NeRF) | Computational Pathology | 3D Reconstruction & Volumetry | Deep Learning | Computer Vision | Transformer Architectures | Natural Language Processing | Healthcare AI

My research is broadly situated in healthcare AI, with a focus on building deep learning systems that are clinically grounded, uncertainty aware, and capable of operating under the constraints real medical data imposes including sparse observations, severe class imbalance, and significant domain shift. I am drawn to problems at the intersection of medical image analysis, 3D reconstruction, neural implicit representations, and computational pathology, where the gap between algorithmic performance and clinical utility remains wide and consequential. My broader objective is to develop models that generalize robustly across imaging modalities and acquisition conditions, that produce outputs clinicians can meaningfully act on, and that hold up under the rigorous evaluation standards medical deployment demands, spanning the full pipeline from physics-informed reconstruction and segmentation to uncertainty quantification and domain adaptation.

Publications

Sparse-View Lung Nodule Volumetry via AReT: Anatomy-Regularized TensorRF | *Under Review*

[arXiv:2606.02639](https://arxiv.org/abs/2606.02639)

2026

- M Spoorthi, P Suja. Manuscript under review.
- Identified and resolved an unreported TensorRF failure mode — default density shift $\Delta = -10$ suppresses all density gradients via dead ReLU throughout training — by setting $\Delta = 0$, restoring gradient flow with zero architectural changes, enabling sparse-view X-ray attenuation reconstruction for the first time in TensorRF.
- Proposed AReT, an anatomy-regularized TensorRF framework for 3D lung nodule volumetry from only three orthogonal digitally reconstructed radiographs (coronal, sagittal, axial); achieved Pearson $r = 0.983$ ($p < 0.0001$) and median APE 11.4% on clinically actionable nodules ≥ 10 mm ($n = 14$), an $8.4\times$ improvement over spherical approximation.
- Designed thorax-specific $\ell_1 + \text{TV}$ regularization exploiting air-dominant thoracic anatomy as an inductive bias, enabling stable 3D reconstruction from 49,152 training rays against 869,555 parameters; implemented Beer-Lambert DRR simulation and HU-calibrated DICOM preprocessing pipeline in PyTorch on NVIDIA T4 (≈ 8 min/patient).
- Conducted a systematic 11-strategy progressive ablation across MedNeRF variants, CheXNet, MONAI UNet, and TensorRF configurations on LIDC-IDRI ($n = 19$), demonstrating anatomy-regularized prior-free TensorRF outperforms all generative-prior-guided approaches (Wilcoxon $W = 0$, $p < 0.0001$).

AMN: Adaptive Multi-Scale Fusion Network for Nuclei Segmentation | *Under Review*

2026

- M Spoorthi, P Suja. Manuscript under review.
- Proposed AMN, a dual-encoder segmentation framework pairing a Swin Transformer and ResNet-50 feature pyramid,

fused at four spatial scales via a learned per-channel sigmoid gate $\alpha \in [0, 1]^{256}$, dynamically blending global context and local texture for seven-class nuclei segmentation in H&E histopathology images.

- Designed a multi-task prediction head producing simultaneous binary segmentation, boundary detection, seven-class classification, and heteroscedastic uncertainty outputs, trained with a composite loss of class-weighted focal loss ($\gamma=2.0$), boundary-aware BCE (positive weight $10\times$), and uncertainty-modulated cross-entropy; achieved mean F1 of 0.6869 and $F1_{Ly}$ of 0.8187 on CoNIC (+12.6 points over ResU-Net).
- Benchmarked against six reimplemented baselines (U-Net, ResU-Net, DeepLabV3+, SegNet, ViT-Small, StarDist) on CoNIC (4,981 patches, 7 classes) with zero-shot cross-dataset evaluation on MoNuSeg (14 images, 7 organs), achieving highest binary F1 without retraining.
- Addressed severe class imbalance (background 83.87%, rare classes $<1\%$) via per-class focal weights, weighted random sampling, and boundary-aware supervision; trained with AdamW, cosine annealing, 5-epoch linear warmup, and gradient accumulation (effective batch 64) for up to 200 epochs with early stopping.

Addressing Unresolved Crimes in India: AI-Driven Insights for Effective Policymaking | *Under Review* 2026

- S Anirudh, G Angelina, M Spoorthi, CR Kavitha. Manuscript under review. Performs exploratory data analysis on 40,160 unresolved crime records across Indian cities, employing Falcon-7B-Instruct for automated crime categorization and spatiotemporal visualization for law enforcement policymaking.

Evaluating BERT-Based Transformer Models for Improved Phishing Detection Using NLP Techniques | *Under Review* 2026

- S Anirudh, M Spoorthi, CR Kavitha. Manuscript under review. Benchmarks BERT, RoBERTa, DistilBERT, and XLNet on phishing email classification with a four-strategy augmentation pipeline; RoBERTa achieved the highest accuracy (96.65%) and F1-score (96.64%).

KannaDa – A Kannada-based Programming Language Interpreter using Lex and Yacc | *IEEE*

[IEEE Xplore](#)

Jun. 2025

- M Spoorthi, P Manaswini, S Shaik, CR Kavitha. Published in IEEE, June 2025. Develops a Kannada-syntax programming language interpreter using compiler design principles and NLP techniques to democratize coding for native Kannada speakers.

Advancing Supply Chain Management: Cloud-Based Demand Forecasting Solutions | *IEEE*

[DOI: 10.1109/ICONAT61936.2024.10775167](#)

Dec. 2024

- M Spoorthi, SS Rao, S Mishra, SM Rajagopal, M Prashanth. Published in IEEE ICONAT 2024. Explores AWS-based demand forecasting for supply chain management using AWS Forecast and ML algorithms for scalable inventory optimization.

Harnessing the Power of Web3: A Blockchain Approach to Crowdfunding | *IEEE*

[IEEE Xplore](#)

Jul. 2024

- RV Savant, SN Sunder, M Spoorthi, SM Rajagopal, RM Balakrishnan. Published in IEEE, July 2024. Proposes a decentralized crowdfunding application using Ethereum, Solidity smart contracts, and MetaMask wallets to enable secure, transparent peer-to-peer fundraising.

A Comparative Analysis of Black-Box and White-Box Models for IoT Botnet Detection | *IEEE*

[DOI: 10.1109/i2ct61223.2024.10544270](#)

May 2024

- RV Savant, M Spoorthi, PK Mohan, S Bhaskaran, SM Rajagopal. Published in IEEE I2CT 2024. Employs a two-stage binary and multi-class ML classification pipeline on CTU-13 and N-BaIoT datasets; LGBM and XGBoost demonstrated superior botnet detection performance.

Unveiling Hidden Patterns: Clustering Algorithms on C Code Embedding | *IEEE*

[IEEE Xplore](#)

May 2024

- M Spoorthi, RV Savant, S Seshadri, N Narmada, PB Pati. Published in IEEE, May 2024. Applies ML clustering (K-means, GMMs) to CodeBERT embeddings of C programs to uncover structural patterns, highlighting the limits of embedding-based automated code grading.

Additional Projects

Vision-Based Framework for Autonomous Navigation in Agricultural Domains |

Amrita Vishwa Vidyapeetham

Nov. 2024 – Mar. 2025

- Designed a fully vision-based autonomous navigation framework for agricultural and warehouse environments using ZED X stereo cameras, vSLAM, and the Nav2 framework, eliminating dependency on LiDAR sensors.
- Integrated stereo vision, IMU fusion, GNSS localization, waypoint navigation, and recovery fallback mechanisms for robust real-time localization, 3D mapping, obstacle avoidance, and adaptive path planning in unstructured terrains.
- Extended the framework to support crop monitoring, sprayer integration, and human detection for smart farming applications.

Advanced Topic Modeling for Categorizing Software Vulnerabilities |

Amrita Vishwa Vidyapeetham

Sep. 2024 – Nov. 2024

- Investigated advanced topic modeling approaches and LLMs (BERTopic with multiple configurations, CombinedTM, Top2Vec, Mixtral, Llama2) to extract semantic patterns from software vulnerability data.
- Applied dimensionality reduction and clustering methods to enhance interpretability and coherence of extracted vulnerability topics for cybersecurity professionals.
- Implemented the full pipeline using Hadoop and PySpark for scalable big data processing of large vulnerability datasets.

BERT-Based Transformer Models for Phishing Email Detection |

Amrita Vishwa Vidyapeetham

Jan. 2026 – Mar. 2026

- Designed and evaluated an intelligent phishing email detection system using transformer-based NLP architectures including BERT, RoBERTa, DistilBERT, and XLNet with comprehensive comparative evaluation.
- Implemented multiple data augmentation strategies (text paraphrasing, syntax transformation, semantic noise injection, character-level perturbations) to improve robustness against evolving phishing techniques.

Experience

Accenture

Oct. 2025 – Present

Advanced Application Engineering Analyst

Bengaluru, India

- Maintain and optimize CI/CD pipelines in Azure DevOps, automating build, test, and deployment workflows across cloud environments on Microsoft Azure and AWS.
- Author smoke testing scripts in Pester (PowerShell) to validate post-deployment service health, and manage source control and release automation via GitHub Actions.

GE Aerospace

Jan. 2025 – Jun. 2025

Digital Technology Intern

Bengaluru, India

- Built and optimized RESTful APIs using Java (Spring Boot) with JPA and PostgreSQL for backend services, ensuring quality through JUnit, Postman, and code reviews.
- Containerized microservices with Docker for deployment via AWS CodePipeline.

**Centre for Development of Telematics (C-DOT), Telecom R&D Institute,
Govt. of India**

Jun. 2024 – Jul. 2024

Internship Trainee – Network Management

Bengaluru, India

- Developed a comprehensive web application for real-time network monitoring, performance analysis, and failure reporting, dynamically retrieving and visualizing data from a MySQL database.
- Built modules for task execution log tracking, system status monitoring, failure rate analysis, and root cause identification using Java EE, JDBC, AngularJS, and MySQL.

**Centre for Development of Telematics (C-DOT), Telecom R&D Institute,
Govt. of India**

Jul. 2023 – Aug. 2023

Internship Trainee – Data Science for 5G O-RAN

Bengaluru, India

- Developed machine learning solutions for 5G O-RAN use cases including context-based dynamic handover management for V2X communications, QoE/QoS optimization, local indoor positioning on RAN, and energy saving using Python and ML libraries.

Technical Skills

Languages: Python, Java, SQL, C++

Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, HuggingFace, Transformers, PySpark

Tools & Platforms: AWS, Docker, Git, Linux, Hadoop, PostgreSQL, MySQL

Leadership / Extracurricular

National Student Data Corp (NSDC) – Bengaluru Chapter

Jan. 2024 – Aug. 2025

Co-Founder

Amrita Vishwa Vidyapeetham, Bengaluru

- Co-founded the Bengaluru chapter of the National Student Data Corp to foster data science and machine learning culture among students at Amrita School of Computing.

Forum for Aspiring Computer Engineers (FACE Amrita)

Jan. 2023 – Jan. 2024

Senior Executive

Amrita Vishwa Vidyapeetham, Bengaluru

- Organized hackathons, technical workshops, and tutorial sessions across various branches of computer science, fostering peer learning and practical skill development within the department.